

Forecasting Fundamentals

Module 6 · Point forecasts, prediction intervals, and loss functions

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Where we are: five modules, one chain

Module 1

Time series have memory

The past is informative about the present. ϕ measures how long a shock lasts.



Modules 1–2

Stationarity

A verdict built from evidence, not a measurement: the plot, the ACF, ADF with KPSS, the rule of thumb.



Module 3

The ACF and PACF are the keys to the process

Every ARMA class leaves a fingerprint: which one tails off, which one cuts off.



Module 4

Model selection criteria

When both tail off, information criteria score the candidates: fit minus penalty.



Modules 4–5

Box-Jenkins general-to-specific ARMA modeling

A procedure, not a pick. Start one size bigger, check the residuals, then trim.



Module 5

Serial correlation

A property of the fitted model, not of the data. Residuals e_t should behave like innovations ϵ_t .

$$y_t = \alpha + \delta t + \sum_{j=1}^p \phi_j y_{t-j} + \sum_{l=1}^q \theta_l \epsilon_{t-l} + \epsilon_t$$

Every knob on the mixing board is on. Module 6 adds no new slider: it uses the fitted equation to project forward.

Before we forecast anything: the whole chain, run once, on one real series.

Know your data: real GDP per capita

Real gross domestic product per capita

FRED [A939RX0Q048SBEA](#) · chained (2017) dollars per person · as published September 30, 2026

$$\text{real GDP per capita} = \frac{\text{real GDP, chained (2017) dollars}}{\text{midperiod population}}$$

WHO PUBLISHES IT

Bureau of Economic Analysis

NIPA Table 7.1, line 10.

Numerator: Table 1.1.6, line 1.

Denominator: Table 7.1, line 18.

FREQUENCY

Quarterly

Quarterly from 1947 Q1; annual from 1929.

Here: 1990 Q1 to 2006 Q4, 68 quarters.

POPULATION

Census Bureau estimates

Total population, including Armed Forces overseas and the institutionalized population.

Quarterly figure: the average of the months.

DEFLATOR

No single deflator

Built bottom-up. Most detailed components: BLS price indexes (CPI, PPI, International Price Indexes).

Aggregated with a Fisher chain-type index. The implicit price deflator is a by-product.

REVISIONS

Advance, second, third

An annual update each September; a comprehensive update about every five years.

Population revises on its own calendar.

SEASONAL ADJUSTMENT

Seasonally adjusted at annual rates

Indirect: components are adjusted, then aggregated.

Population is not seasonally adjusted.

ACCOUNTING

An aggregate

$$GDP = C + I + G + (X - M)$$

$$GDP = GDI + \text{statistical discrepancy}$$

Chained-dollar components are not additive: the tables carry a “Residual” line.

GUIDING DOCUMENTS

SNA 2008 · NIPA Handbook

The NIPAs are “generally consistent with the SNA”.

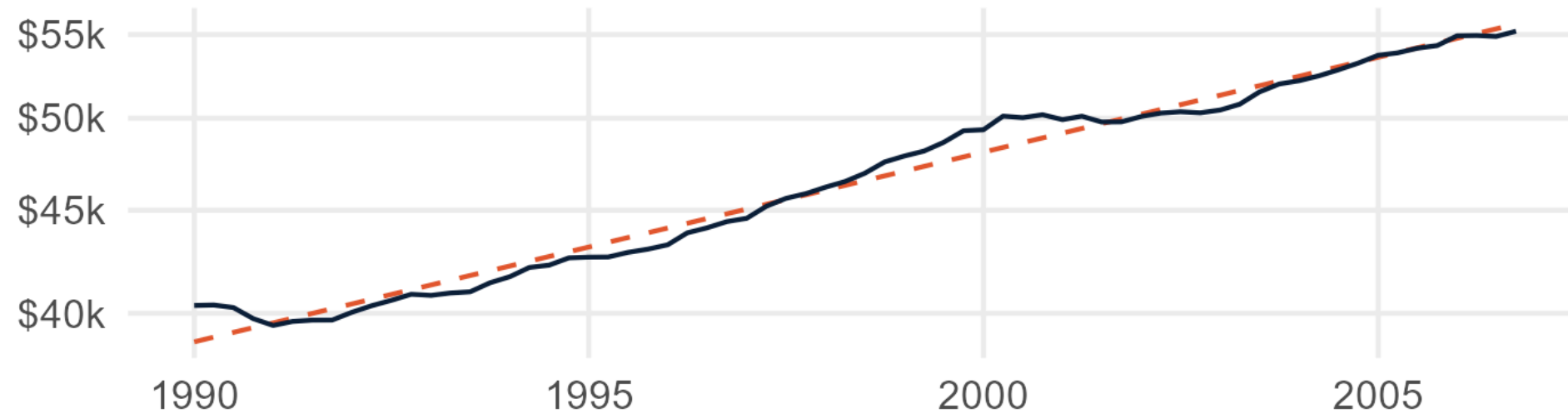
BEA: *Concepts and Methods of the U.S. National Income and Product Accounts*.

A ratio of two estimates. Both get revised; only one is seasonally adjusted.

The chain starts: is it stationary?

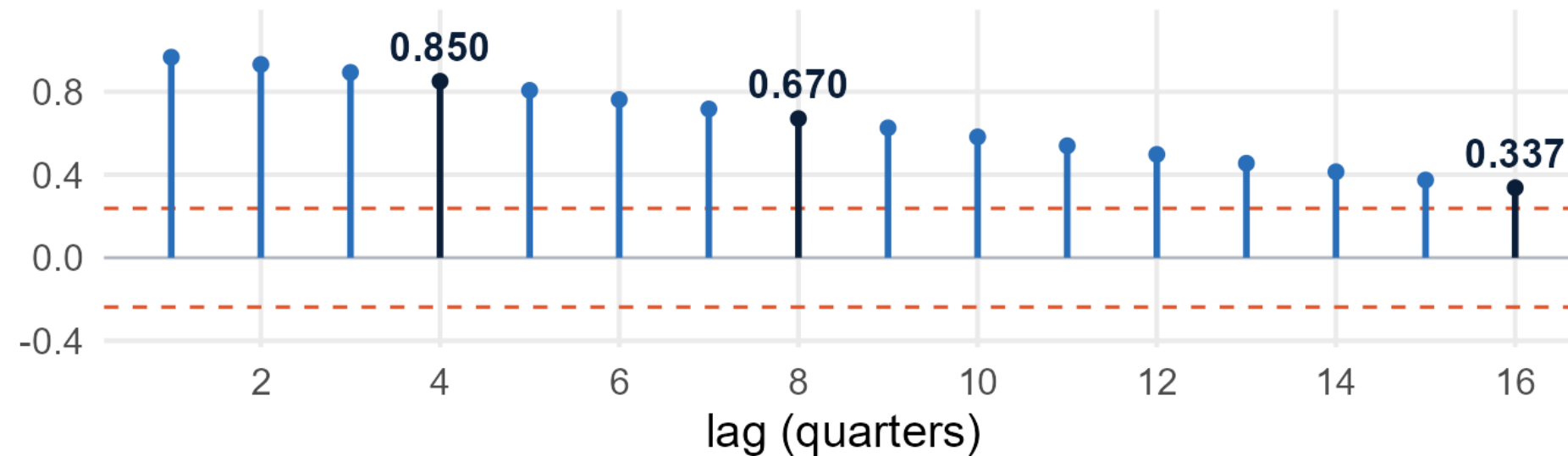
Log real GDP per capita, 1990 Q1 to 2006 Q4

natural log scale, dollar labels; dashed: linear trend, 2.17% a year



Sample ACF of the log level

no model fitted: the series' own autocorrelation; band = $\pm 1.96/\sqrt{T}$



y_t is the natural log of real GDP per capita. Why logs: constant growth is a straight line, and Δy_t is the quarterly growth rate. The fitted trend grows 2.17% a year.

ADF · trend H_0 : unit root

2 lags, chosen by BIC

Alternative: stationary around a trend

τ against the 5% critical value **-3.11 vs -3.45**

Not below the critical value: **fail to reject a unit root**

KPSS · trend H_0 : stationary around a trend

Statistic, short truncation **0.150 vs 0.146**

Statistic, long truncation **0.085 vs 0.146**

Above by a hair with one truncation, below with the other: **no clear rejection**

Rule of thumb No null: a persistence flag

$R = \sigma_{\Delta y} / \sigma_y$ against the threshold **0.050 vs 0.5**

Far below 0.5, but the trend itself inflates σ_y : this row cannot separate the two readings by itself

Neither null is clearly rejected. Can 68 quarters tell a trend from a unit root?

Trend-stationary, or a unit root?

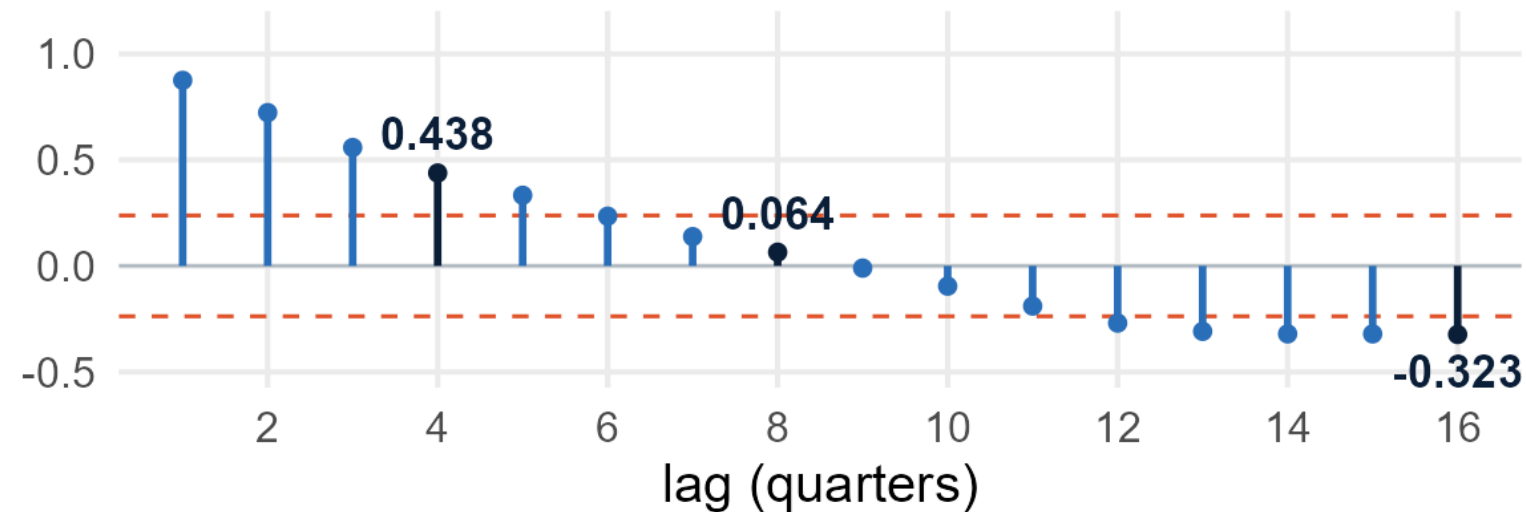
Deviations from the linear trend

100 x (log y minus fitted trend), percent; 1990 Q1 to 2006 Q4



Sample ACF of the deviations

their own autocorrelation; band = $\pm 1.96/\sqrt{T}$



Deviations How persistent are they?

Rule of thumb on the deviations, $\sigma_{\Delta y}/\sigma_{\hat{u}}$ vs 0.5

0.359

AR(1) coefficient of $\hat{u}_t$ (descriptive)

0.879

Half-life of a deviation (descriptive)

5.4 quarters

Robustness Rejected at 5%? Lags 0, 1, 2, 4, 8

Unit root, ADF: τ from -3.11 to -1.95 vs -3.45

no

Unit root, DF-GLS: from -2.11 to -1.39 vs -3.03

no

Trend-stationarity, KPSS: 0.150 short, 0.085 long vs 0.146

split

No lag choice rejects a unit root. With 68 quarters that is low power (Module 2: near the boundary these tests rarely reject), not evidence of a unit root. KPSS sits on its critical value. The deviations are persistent, yet keep returning to the line: the borderline case.

A choice, not a finding. 68 quarters cannot separate the two readings.

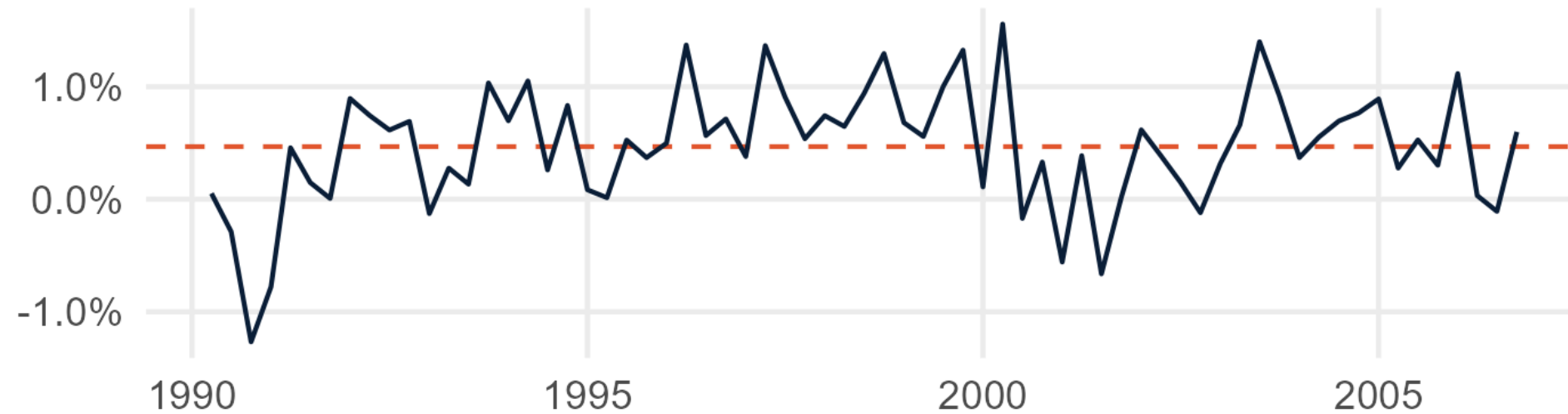
The course's rule under doubt is to difference: treat y_t as $I(1)$, model the growth rate Δy_t , then check that fit for over-differencing.

Why it matters for forecasting: under a deterministic trend the forecast returns to the line and the interval stops widening; under a unit root it does not.

Was the first difference enough?

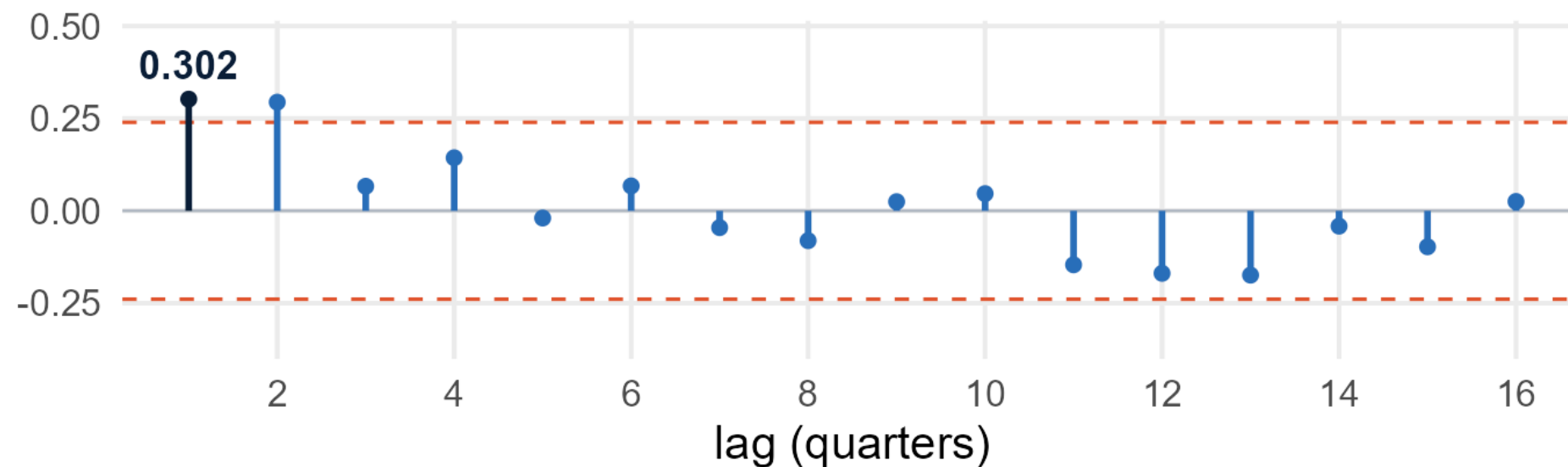
Quarterly growth, 1990 Q2 to 2006 Q4

100 x first difference of log y, percent; dashed: mean, 0.47%



Sample ACF of the first difference

the series' own autocorrelation; band = $\pm 1.96/\sqrt{T}$



ADF · drift H_0 : unit root

1 lag, chosen by BIC

τ against the 5% critical value

-3.83 vs -2.89

Below the critical value: **reject a unit root**

One lag choice disagrees. At 0, 1, 2 and 4 lags ADF and DF-GLS both reject; at 8 lags neither does (ADF -2.16 vs -2.89).

KPSS · level H_0 : stationary around a constant mean

Statistic, short truncation

0.214 vs 0.463

Statistic, long truncation

0.166 vs 0.463

Below the critical value with both: **fail to reject stationarity**

Rule of thumb No null: a persistence flag

$R = \sigma_{\Delta^2 y} / \sigma_{\Delta y}$ against the threshold

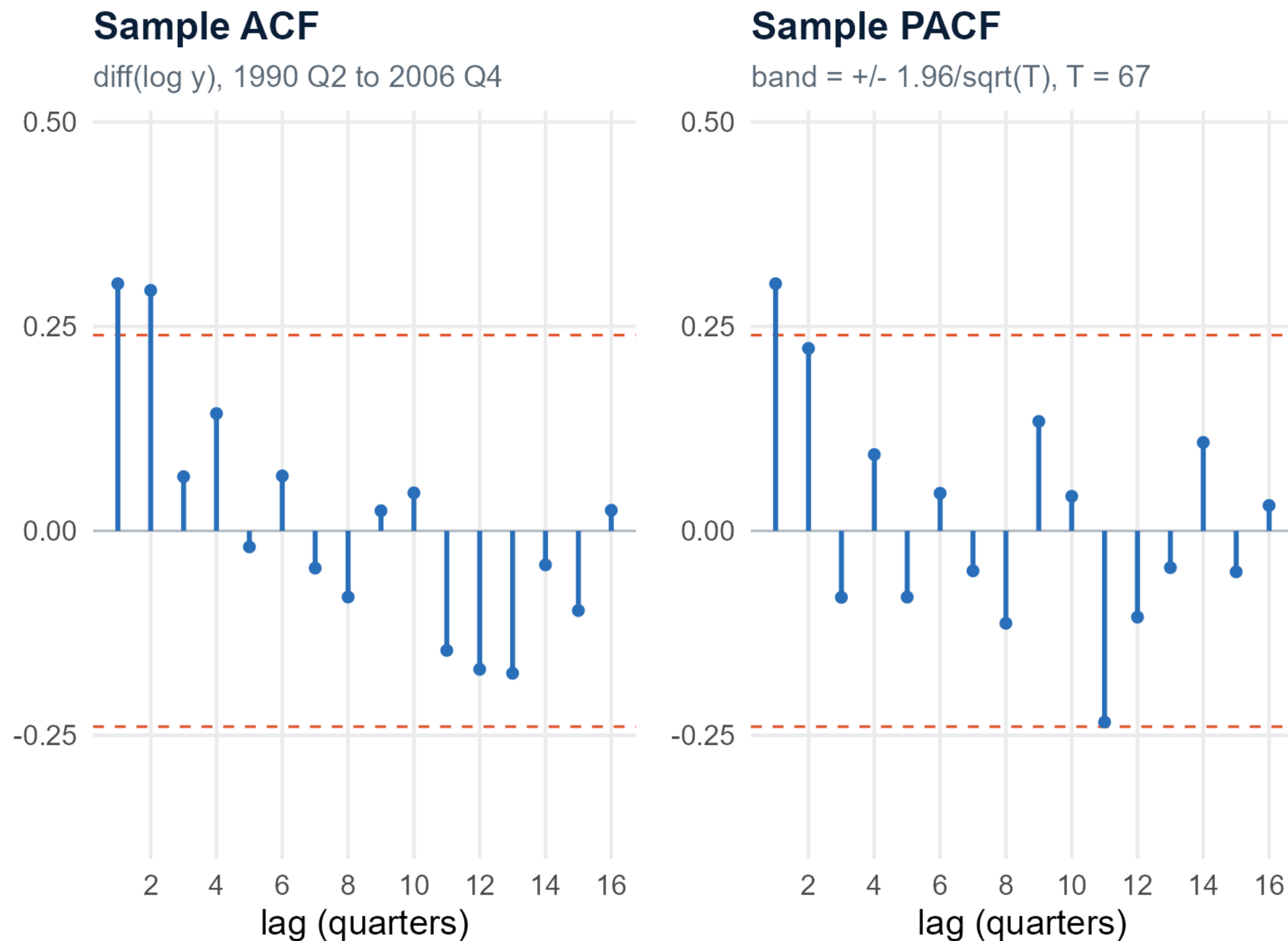
1.186 vs 0.5

Above 0.5: no second difference needed. **Keep Δy_t**

One difference is enough; the only dissent is the 8-lag test. Δy_t is the series we model.

No sign of over-differencing so far: $\hat{\rho}_1 = 0.30$, where an MA(1) with $\theta = -1$ would give -0.5 . The real check is $\hat{\theta}$ near -1 once we fit.

The series we model: ACF and PACF of Δy_t



Both tail off. The ACF clears the band at lags 1 and 2; the PACF at lag 1, with lag 2 (0.22) just inside; then both fade. The ACF decays no faster than the PACF.

Module 3's table: ACF tails off, PACF cuts off, means AR; both tail off means ARMA. Nothing here cuts off cleanly, so AR or ARMA, not MA.

The general choice is ARMA: $\text{ARMA}(p, q)$ nests $\text{AR}(p)$ by setting the θ 's to zero, so it costs parameters, not possibilities.

Start one size bigger, then trim (Module 5's rule): fit the larger model and let the criteria and the residuals cut it down.

Starting model: $\text{ARIMA}(2,1,2)$ with drift

$\text{ARMA}(2,2)$ on Δy_t with a constant. The constant is the drift: growth averages 0.47% a quarter, and a model without it would forecast zero growth.

Information criteria: one scoreboard, four tariffs

$$IC = -2 \log L(\hat{\boldsymbol{\vartheta}}) + \text{penalty}(k, T)$$

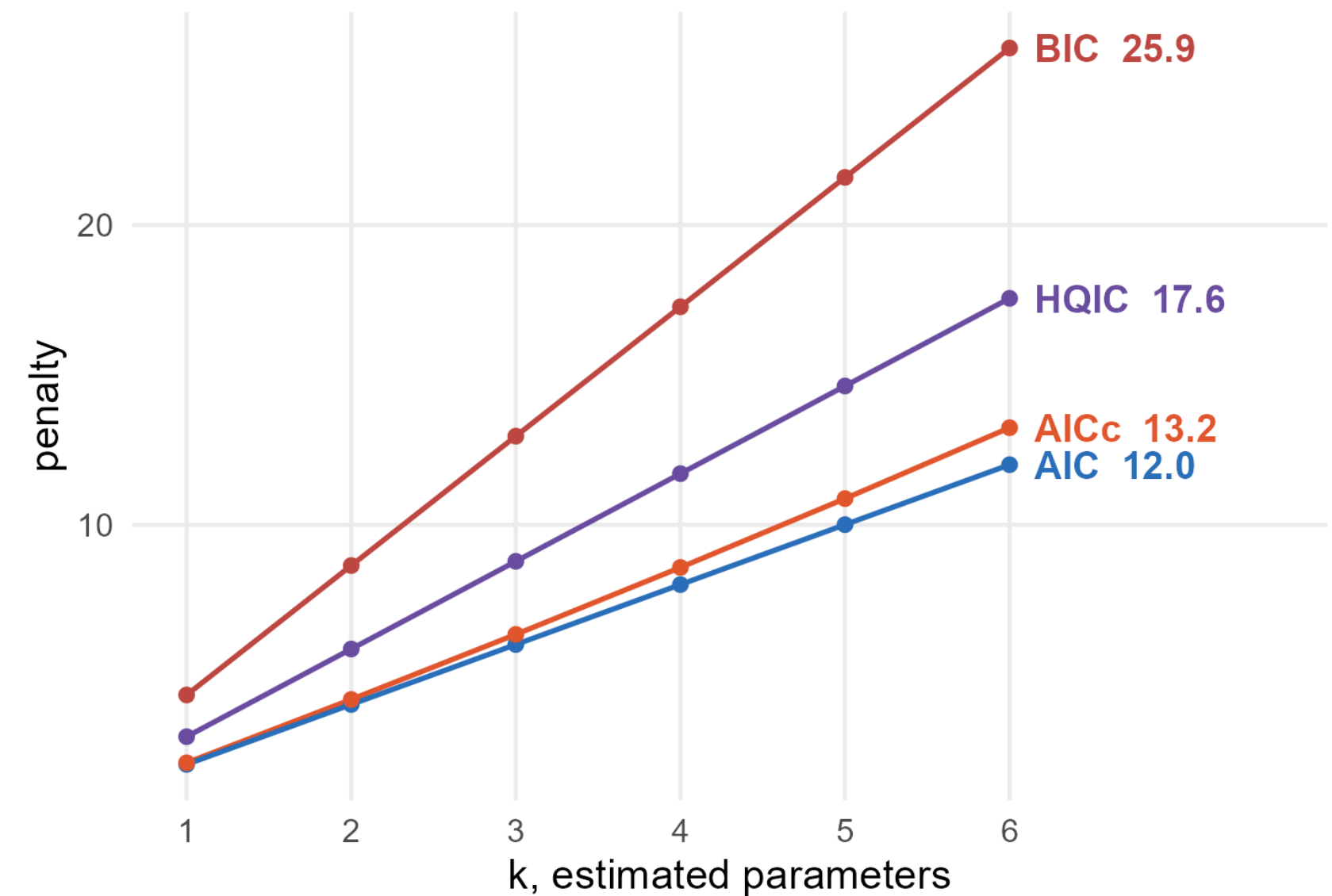
Fit first: a bigger log-likelihood is a smaller deviance. Then the tariff for complexity. **Smaller IC is better.**

AIC (Akaike, 1974)	$2k$
AICc (Hurvich and Tsai, 1989)	$2k + \frac{2k(k+1)}{T-k-1}$
BIC (Schwarz, 1978)	$k \log T$
HQIC (Hannan and Quinn, 1979)	$2k \log \log T$

k counts every estimated parameter: the AR and MA coefficients, the drift, and σ^2 . T is the number of observations the fit uses.

The penalty term at $T = 75$

what each criterion adds to $-2 \log L$ for k estimated parameters



Per parameter at $T = 75$: BIC $\log 75 = 4.32$, HQIC $2 \log \log 75 = 2.93$, AIC 2; AICc adds a small-sample term. BIC punishes parameters hardest at this T , so it favours the smallest models.

The candidates, scored

```

1 library(forecast)
2 source("helpers/model_selection.R")      # hqic()
3 y    <- log(gdp_pc)                      # 1990 Q1 to 2006 Q4
4 fit <- function(p, q)
5   Arima(y, order = c(p, 1, q), include.drift = TRUE)
6
7 fits <- list(m212 = fit(2, 2), m211 = fit(2, 1),
8             m111 = fit(1, 1))
9 sapply(fits, function(f)
10   c(k = length(coef(f)) + 1, logLik = f$loglik,
11     AIC = f$aic, AICc = f$aicc, BIC = f$bic,
12     HQIC = hqic(f)))
13
14 for (ic in c("aicc", "aic", "bic"))      # each minimiser
15   print(auto.arima(y, d = 1, seasonal = FALSE,
16                   stepwise = FALSE,
17                   approximation = FALSE, ic = ic))
18
19 grid <- expand.grid(p = 0:5, q = 0:5)
20 grid <- grid[grid$p + grid$q <= 5, ]    # its grid
21 grid$HQIC <- mapply(function(p, q) hqic(fit(p, q)),
22                    grid$p, grid$q)
23 grid[which.min(grid$HQIC), ]           # HQIC by hand

```

Model	k	$\log L$	AIC	AICc	BIC	HQIC
ARIMA(2,1,2)	6	261.62	-511.25	-509.85	-498.02	-506.01
ARIMA(2,1,1)	5	261.28	-512.56	-511.58	-501.54	-508.20
ARIMA(1,1,1)	4	260.28	-512.56	-511.92	-503.74	-509.07
ARIMA(2,1,0) †	4	260.82	-513.63	-512.99	-504.81	-510.14
ARIMA(1,1,0) †	3	259.00	-512.00	-511.61	-505.38	-509.38

Picks. AIC, AICc and HQIC: ARIMA(2,1,0). BIC: ARIMA(1,1,0). Neither is one of the three we wrote down: † marks the rows `auto.arima` added. All five carry drift.

ARIMA(2,1,2): $\hat{\theta} = (-1.65, 0.65)$, an MA root on the unit circle, Module 3's over-differencing warning light. `auto.arima`'s root screen drops it.

Four criteria, one candidate set: which model, and are its residuals clean?

criterion

AIC

AICc

BIC

HQIC

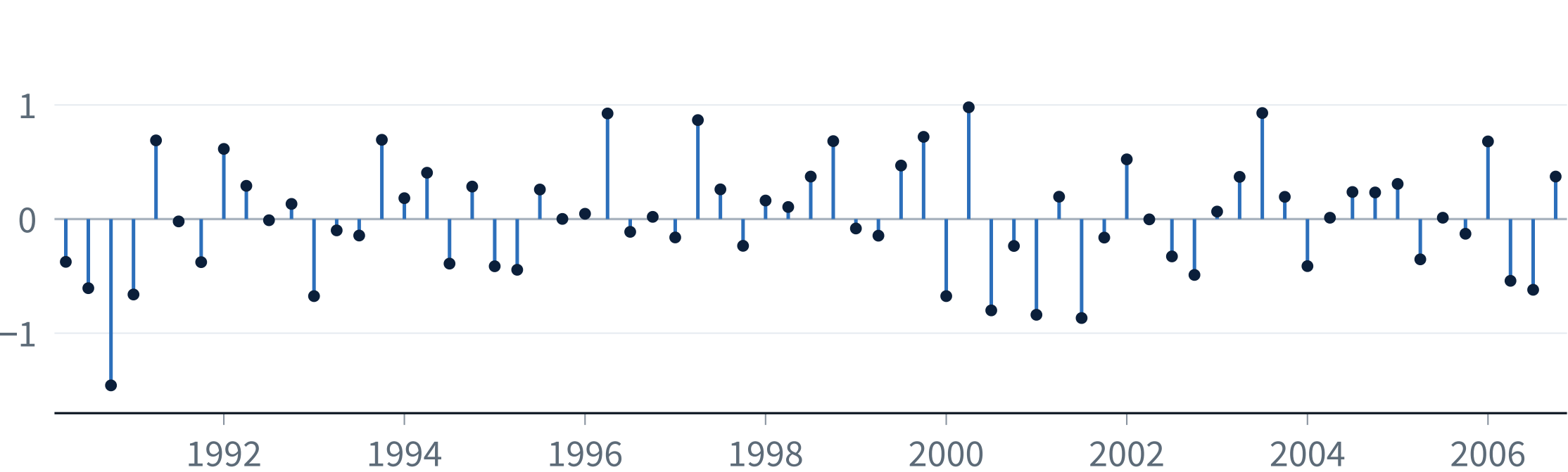
Reset

AICc picks ARIMA(2,1,0) with drift · same model as AIC and HQIC; BIC picks ARIMA(1,1,0) with drift

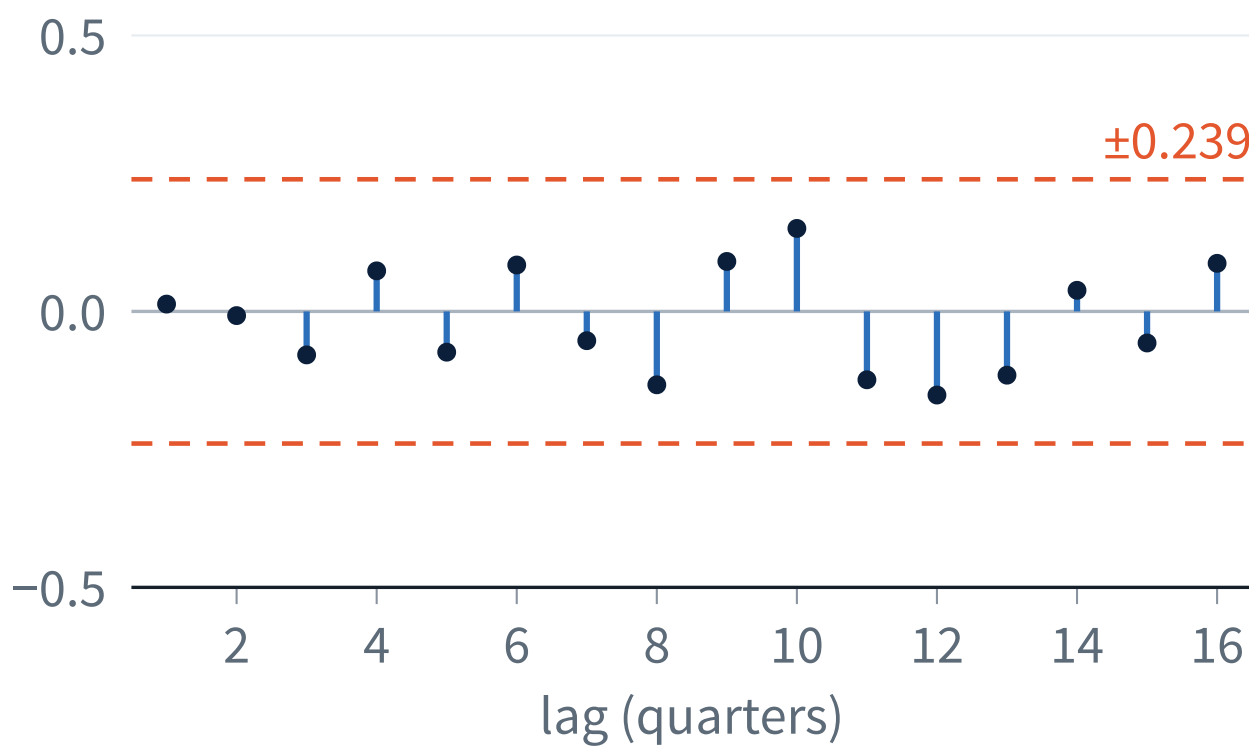
AICc = **-512.99**: $-2 \log L = -521.63$, penalty = 8.65 ($k = 4$, $T = 67$)

$\hat{\phi}_1 = 0.232$ (0.118) · $\hat{\phi}_2 = 0.233$ (0.120) · $\hat{\delta}$ (drift) = 0.0046 (0.0011) · standard errors in parentheses

Residuals e_t of the fitted model, percent per quarter



Sample ACF of the residuals, lags 1 to 16



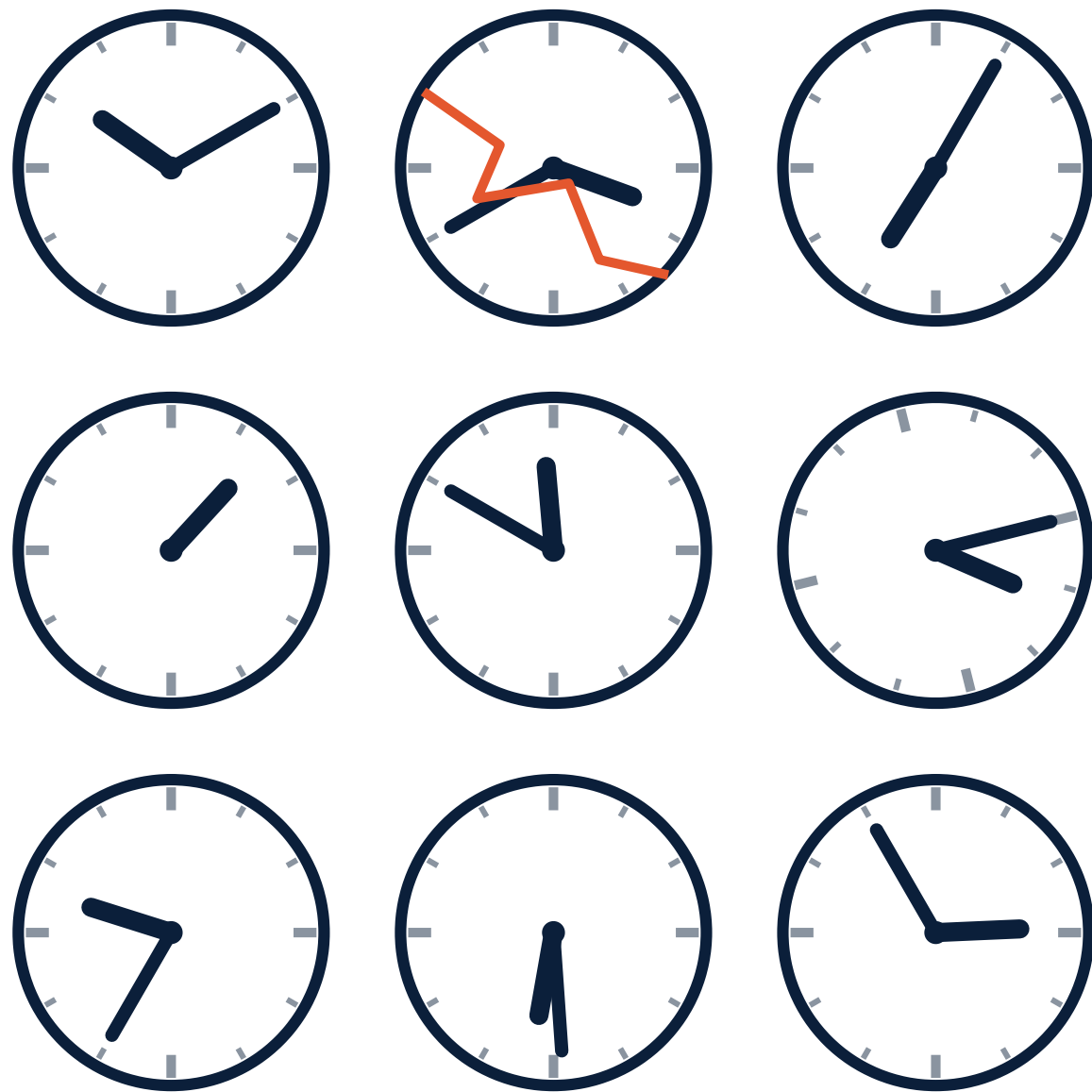
h = 10: $Q = 5.89$, χ^2_8 , $p = 0.660$ **pass** h = 24: $Q = 19.61$, χ^2_{22} , $p = 0.607$ **pass** no visible evidence of remaining serial correlation

AICc over the 5 candidates all with drift; smallest wins	ARIMA(2,1,2) -509.85	ARIMA(2,1,1) -511.58	ARIMA(1,1,1) -511.92	ARIMA(2,1,0) -512.99	ARIMA(1,1,0) -511.61
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Residuals e_t , not innovations ε_t : if the model is right, e_t should behave like ε_t . Serial correlation is a verdict on the fitted model, not on the data. Band $\pm 1.96/\sqrt{T}$, $T = 67$.

You will be wrong

the question is by how much, and whether you said so in advance



Even a stopped clock is right twice a day.

proverb; earliest form Addison, The Spectator no. 129, 28 July 1711

“The moment you forecast you know you’re going to be wrong, you just don’t know when and in which direction.”

from Fiedler’s forecasting rules, *Across the Board*, June 1977

“All models are wrong but some are useful”

George E. P. Box, 1979, p. 202

“Wall Street indexes predicted nine out of the last five recessions!”

Paul Samuelson, *Newsweek*, 19 September 1966

Our sample ends in 2006 Q4. What came next?

“At this juncture, however, the impact on the broader economy and financial markets of the problems in the subprime market seems likely to be contained.”

Ben Bernanke, testimony before the Joint Economic Committee, 28 March 2007

“We simply do not know. Nevertheless, the necessity for action and for decision compels us as practical men to do our best to overlook this awkward fact.”

Keynes, *Quarterly Journal of Economics*, February 1937, p. 214

A forecast has two parts

Origin t where you stand

Horizon h how far ahead you look

Information set $\mathcal{F}_t = \{y_t, y_{t-1}, \dots\}$ everything you know at t

The point forecast. $\hat{y}_{t+h|t} = \mathbb{E}[y_{t+h} | \mathcal{F}_t]$: the best guess under squared-error loss, because the conditional mean minimises expected squared error.

The forecast error. $e_{t+h|t} = y_{t+h} - \hat{y}_{t+h|t}$: unknowable at t , which is the point. It is observable after the fact, so it is an e , not an ϵ .

The prediction interval. $\hat{y}_{t+h|t} \pm c_h$ with a stated coverage, 80% or 95%: the fine print, where the honesty lives.

“Between 3% and 12%”, or “4.1%, give or take 0.3”: which forecast is more likely to be right, and which is more useful?

The headline number is the point forecast; the interval is what keeps it honest. Both are wrong, in different ways: the point in the third decimal, the interval one time in twenty.

The AR(1) point forecast is a recursion

$$y_{t+1} = \alpha + \phi y_t + \epsilon_{t+1}$$

$$\hat{y}_{t+1|t} = \alpha + \phi y_t \quad \text{because} \quad \mathbb{E}[\epsilon_{t+1} | \mathcal{F}_t] = 0$$

$$\hat{y}_{t+h|t} = \alpha + \phi \hat{y}_{t+h-1|t}$$

$$\hat{y}_{t+h|t} - \mu = \phi^h (y_t - \mu) \quad \text{with} \quad \mu = \frac{\alpha}{1 - \phi}$$

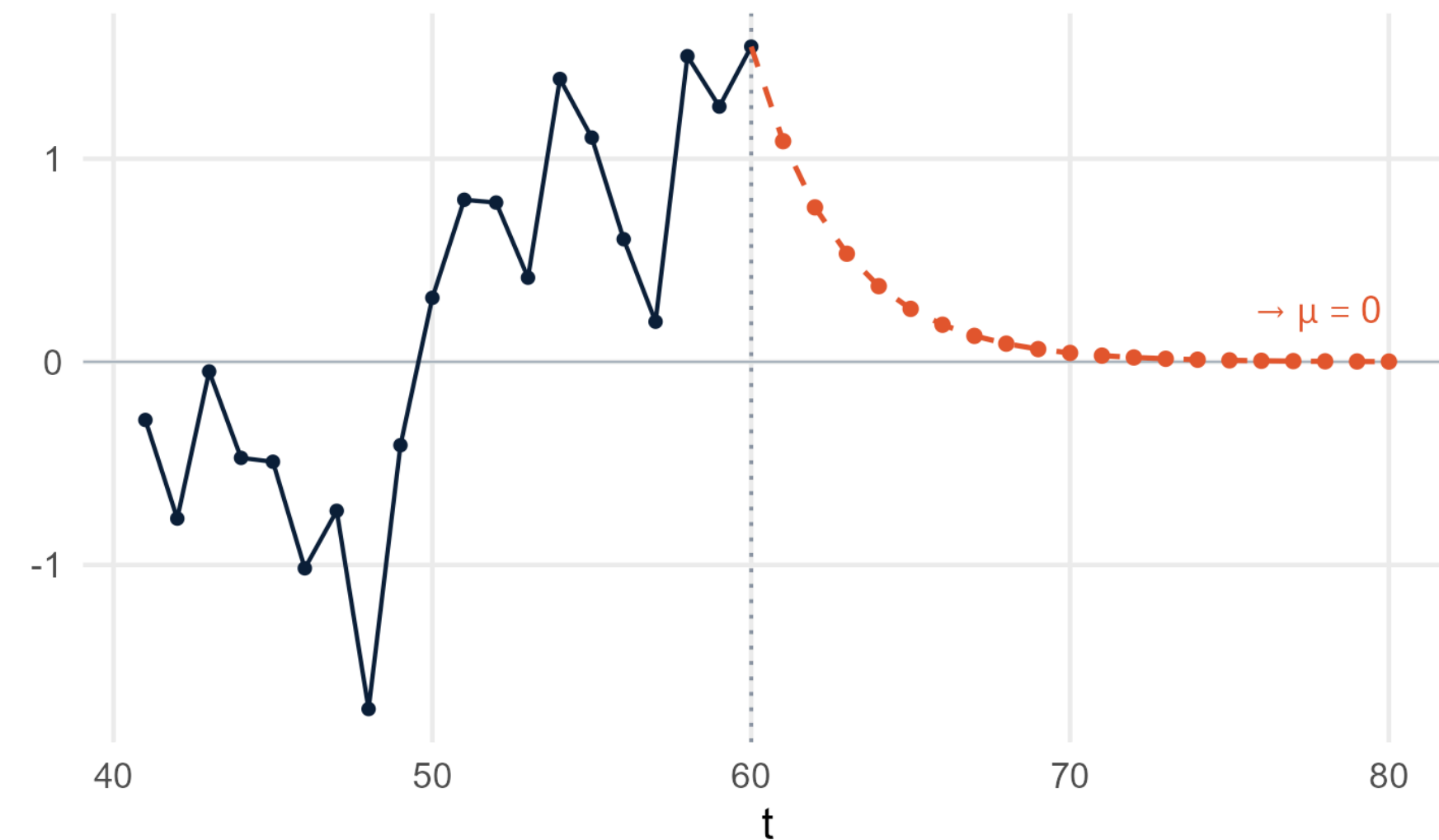
Two rules. Replace future innovations with zero. Replace future y 's with their forecasts. Each step feeds the previous forecast forward: that is why it is called recursive.

R's **intercept** is the mean $\hat{\mu}$, not $\hat{\alpha}$. Recover $\hat{\alpha} = \hat{\mu}(1 - \hat{\phi})$ before you run the recursion, or run it in deviations from $\hat{\mu}$.

As $h \rightarrow \infty$, where does the forecast go?

AR(1), $\phi = 0.7$: the point forecast from $t = 60$

observed (dots) and $\hat{y}_{T+h|T} = \phi^h y_T$ (dashed); $y_T = 1.55$, $\mu = 0$



h	1	2	3	5	20
$\hat{y}_{T+h T} = 0.7^h y_T$	1.086	0.760	0.532	0.261	0.001

Known parameters: $\phi = 0.7$, $\sigma^2 = 1$, $\alpha = \mu = 0$; $T = 60$, origin $y_T = 1.552$. A nonzero μ only shifts everything: the forecast reverts to μ instead of 0.

The AR(1) prediction interval

$$e_{t+h|t} = \sum_{i=0}^{h-1} \phi^i \epsilon_{t+h-i}$$

The shocks you have not seen yet, each weighted by how long the model remembers it. In general $e_{t+h|t} = \sum_{i=0}^{h-1} \psi_i \epsilon_{t+h-i}$ with Module 3's ψ -weights; for an AR(1), $\psi_i = \phi^i$.

$$\text{Var}(e_{t+h|t}) = \sigma^2 \sum_{i=0}^{h-1} \phi^{2i} = \sigma^2 \frac{1 - \phi^{2h}}{1 - \phi^2}$$

At $h = 1$ it is σ^2 . As $h \rightarrow \infty$ it is $\gamma_0 = \sigma^2 / (1 - \phi^2)$: as uncertain as knowing nothing.

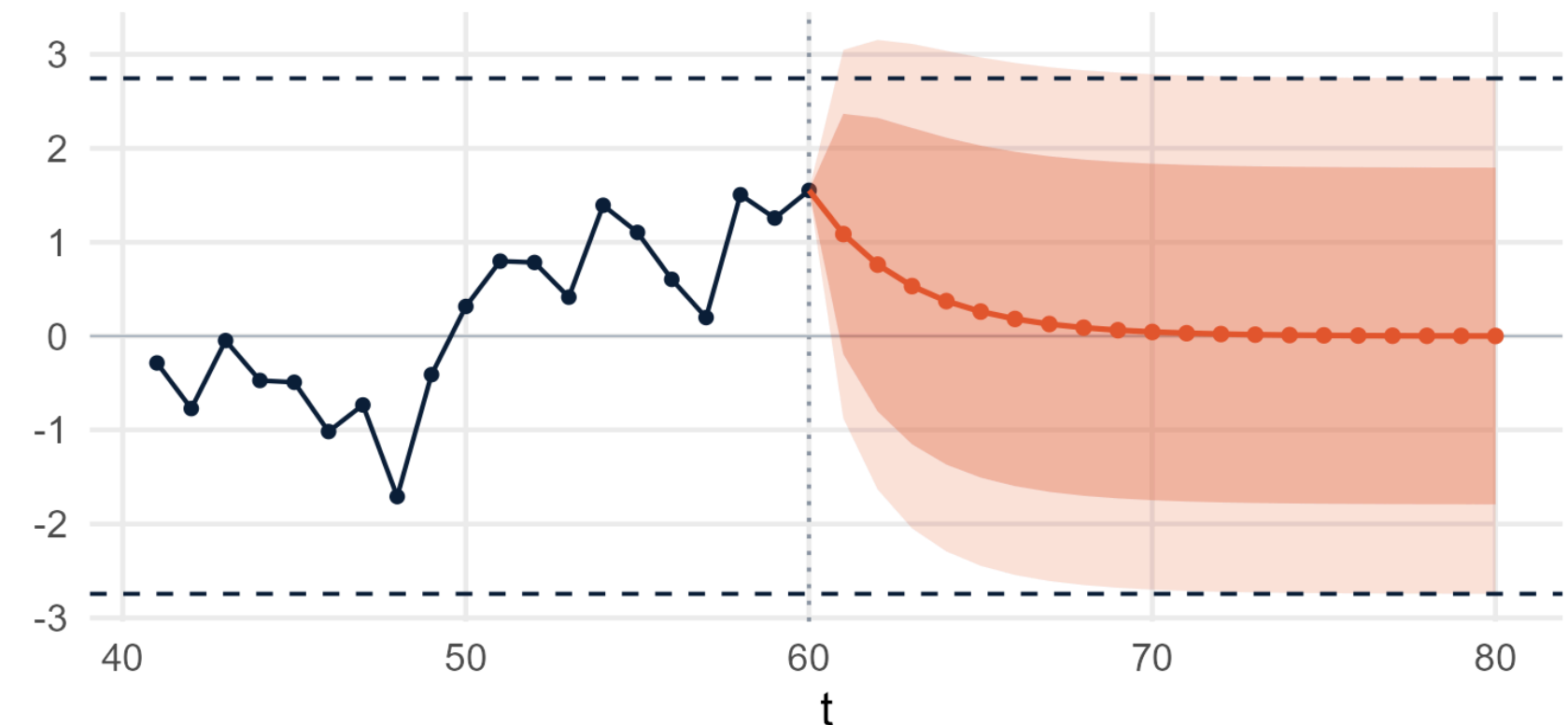
$$\widehat{y}_{t+h|t} \pm z_{\alpha/2} \sigma \sqrt{\frac{1 - \phi^{2h}}{1 - \phi^2}}$$

Gaussian innovations make the error Gaussian, so $z_{\alpha/2} = 1.96$ for a 95% interval and 1.28 for 80%.

The 95% rests on $\epsilon_t \sim N(0, \sigma^2)$, the right model, and a known σ^2 . Break any of them and the interval is nominal, not delivered. The break comes later.

AR(1), $\phi = 0.7$, $\sigma^2 = 1$: the fan

80% and 95% bands; dashed: $\pm 1.96\sqrt{\gamma_0}$, the unconditional limit



$\text{Var}(e_{T+1 T})$	1.000
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$\text{Var}(e_{T+5 T})$	1.905
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$\text{Var}(e_{T+20 T})$ against γ_0	1.961 vs 1.961
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Now move ϕ and σ^2 and watch the fan.

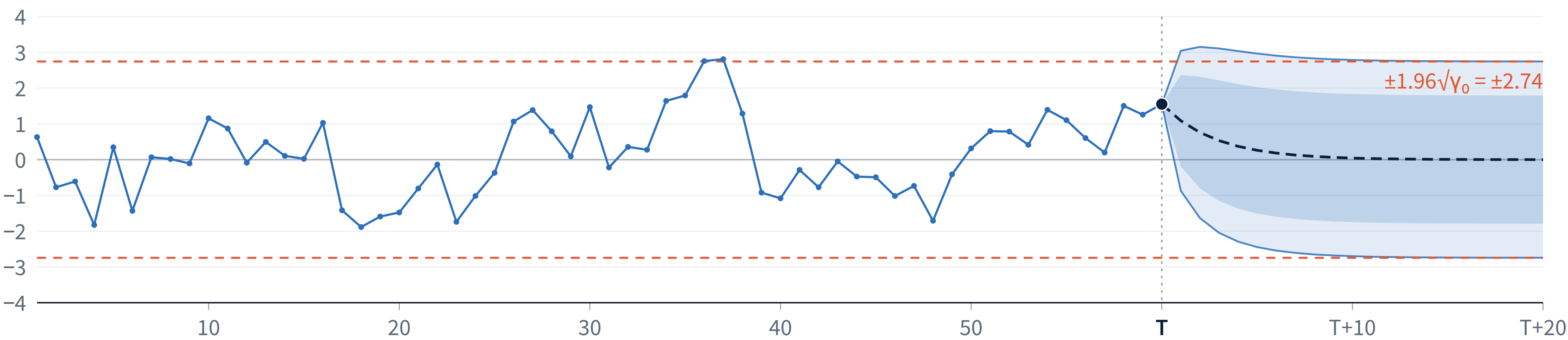
Play it out: an AR(1) forecast with your ϕ and σ^2

model ϕ = 0.70 σ^2 = 1.00 (σ = 1.00) | band ☒ 95% ☐ 80%

play it out | h = 0 of 20 inside the 95% band: 0 of 0

$\hat{y}_{T+h|T} = \phi^h y_T = 0.70^h \times 1.55 \rightarrow h=1: 1.09 \cdot h=20: 0.00$ · decays to the mean, 0, at rate ϕ
 $\text{Var}(e_{T+h|T}) = \sigma^2(1 - \phi^{2h})/(1 - \phi^2) \rightarrow h=1: 1.00 \cdot h=20: 1.96 \cdot \gamma_0 = \sigma^2/(1 - \phi^2) = 1.00/(1 - 0.49) = 1.96$ · 95% band: $\hat{y}_{T+h|T} \pm 1.96\sqrt{\text{Var}}$

—•— observed $y_1 \dots y_{60}$ •-- origin and forecast $\hat{y}_{T+h|T}$ 80% and 95% fans - - - $\pm z\sqrt{\gamma_0}$ unconditional ● inside ● outside the 95% band

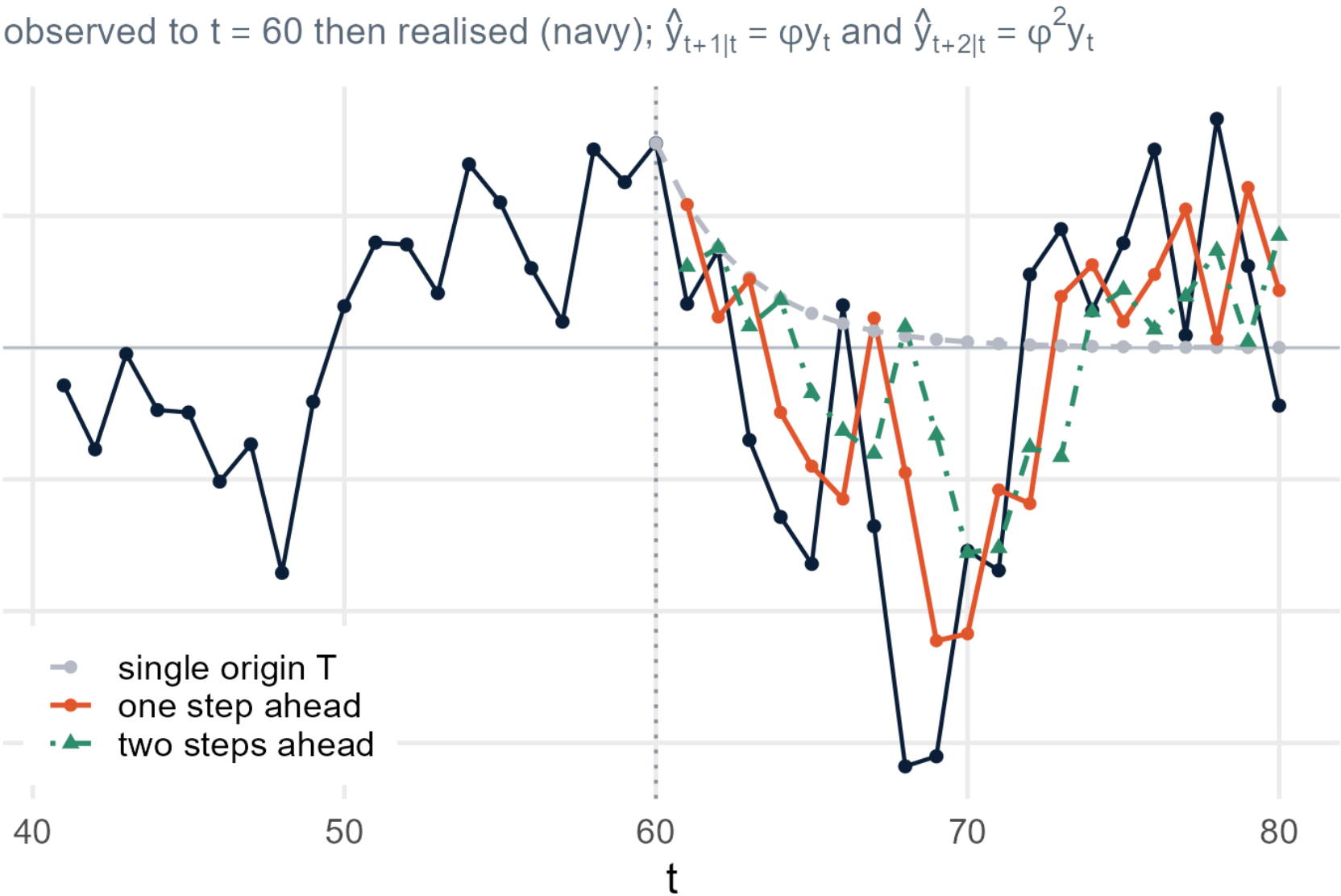


at the origin, $h=0$: $y_T = 1.55$ · first step: $\hat{y}_{T+1|T} = 1.09$, 95% interval $[-0.87, 3.05]$ · press Next step or Play

The fan widens because each step adds one more unresolved shock ε_t ; for a stationary AR(1) it stops at the unconditional band $\pm z\sqrt{\gamma_0}$. The coverage promise (about 95 of 100 realisations inside the 95% band) assumes Gaussian innovations and known ϕ and σ^2 : there is no estimation error here.

Stationary forecasts are short-run forecasts

AR(1), $\phi = 0.7$: rolling one- and two-step forecasts



Same series as slides 13 to 15, $\phi = 0.7$ and $\mu = 0$ known. One-step: $\hat{y}_{t+1|t} = \phi y_t$ from each origin $t = 60, \dots, 79$. Two-step: $\hat{y}_{t+2|t} = \phi^2 y_t$ from $t = 59, \dots, 78$. Grey: the single path from $T = 60$.

Why. $\hat{y}_{t+h|t} - \mu = \phi^h (y_t - \mu)$: the information in y_t decays geometrically, so after a few steps the forecast is μ and the interval is the unconditional one; here the single path is within 0.2 of μ by $h = 6$. The forecast explains the share $\text{Var}(\hat{y}_{t+h|t})/\gamma_0 = \phi^{2h}$ of the variance; the error carries the rest.

at $\phi = 0.7$	$h = 1$	$h = 2$	$h = 4$	$h = 8$
explained, ϕ^{2h}	49%	24%	6%	0.3%
left over, $1 - \phi^{2h}$	51%	76%	94%	99.7%

Persistence is the dial. Half the variance is still explained at $h = \ln 0.5 / (2 \ln \phi)$: about 1 step at $\phi = 0.7$, about 7 at $\phi = 0.95$. ϕ is memory (Module 1).

Reading the picture. One-step forecasts track the series because each uses the latest observation; two-step forecasts track it less; the single path stops tracking after a few steps. Module 7 evaluates forecasts this way: rolling origin, not fixed origin (notes 6.3).

A stationary model forecasts the next few steps; after that it forecasts the mean. ϕ sets how many steps “a few” is.

The example meets its test set: 2007 to 2011

criterion

AIC

AICc

BIC

HQIC

view

growth

level

horizon

h = 1

h = 2

both

benchmark (RW + drift)

band

95%

80%

Next step

Play

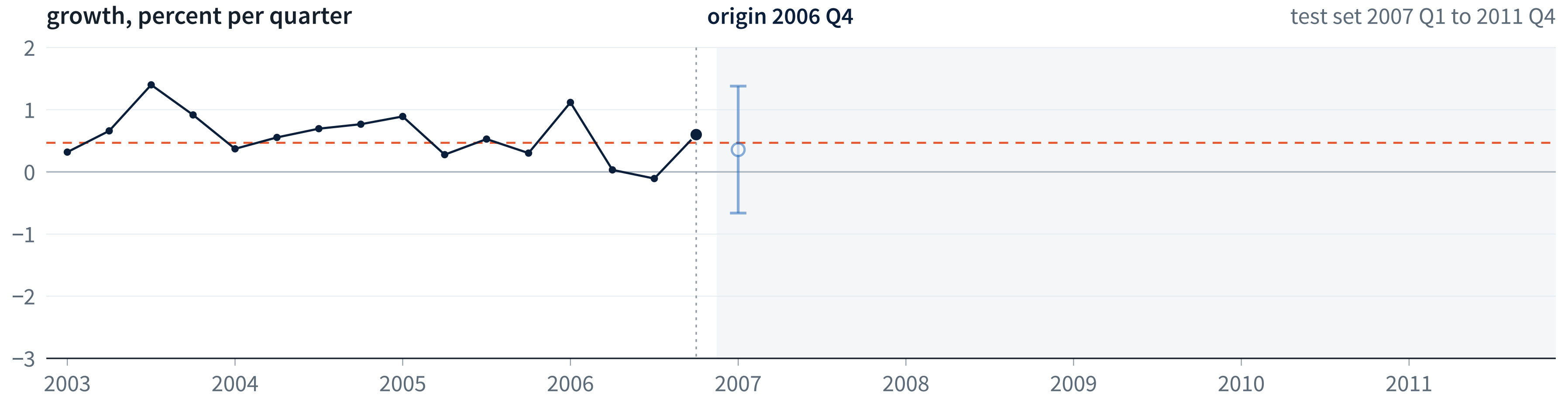
Reset

origin 2006 Q4

inside the 95% band: 0 of 0

AICc picks ARIMA(2,1,0) with drift · same model as AIC and HQIC; BIC picks ARIMA(1,1,0) with drift

—●— realised -○- $\hat{y}$, h = 1 -◇- $\hat{y}$, h = 2 95% band -■- benchmark - - - training drift 0.47 ● inside ● outside



at the origin **2006 Q4**: last observed growth **0.60** · next, 2007 Q1, h = 1 from 2006 Q4: $\hat{y} = 0.36$, 95% band [-0.66, 1.38] · press Next step or Play

Parameters fixed at the 2006 Q4 estimates, no re-estimation as the quarters arrive (Module 7 does that); each quarter's forecast is made h quarters earlier. The data are the 2026-09-30 vintage, not what a forecaster saw in 2007. The intervals assume Gaussian innovations and the right model.

How do we judge ourselves if we are always wrong?

A loss function is the cost of being wrong. $g(e)$, with $e = y_{t+h} - \hat{y}_{t+h|t}$ the forecast error from “A forecast has two parts”. It is g , not L : L is the lag operator.

Zero when the forecast is perfect

$$g(0) = 0$$

Larger as the miss grows

a bigger $|e|$ costs more

Chosen before you see the results, not after. The loss is a decision about what kind of errors hurt most; the choice comes first and the formula follows.

Compare within a loss, never between: same loss, same series, the same N test-set observations.

The Texas grid. You forecast energy demand and a cold snap hits. Under-forecast, and there is not enough supply: people freeze. Over-forecast, and you ran extra generators: wasted fuel. Both are errors; they are not the same error. A symmetric loss says 500 MW under and 500 MW over are equally bad. They are not.

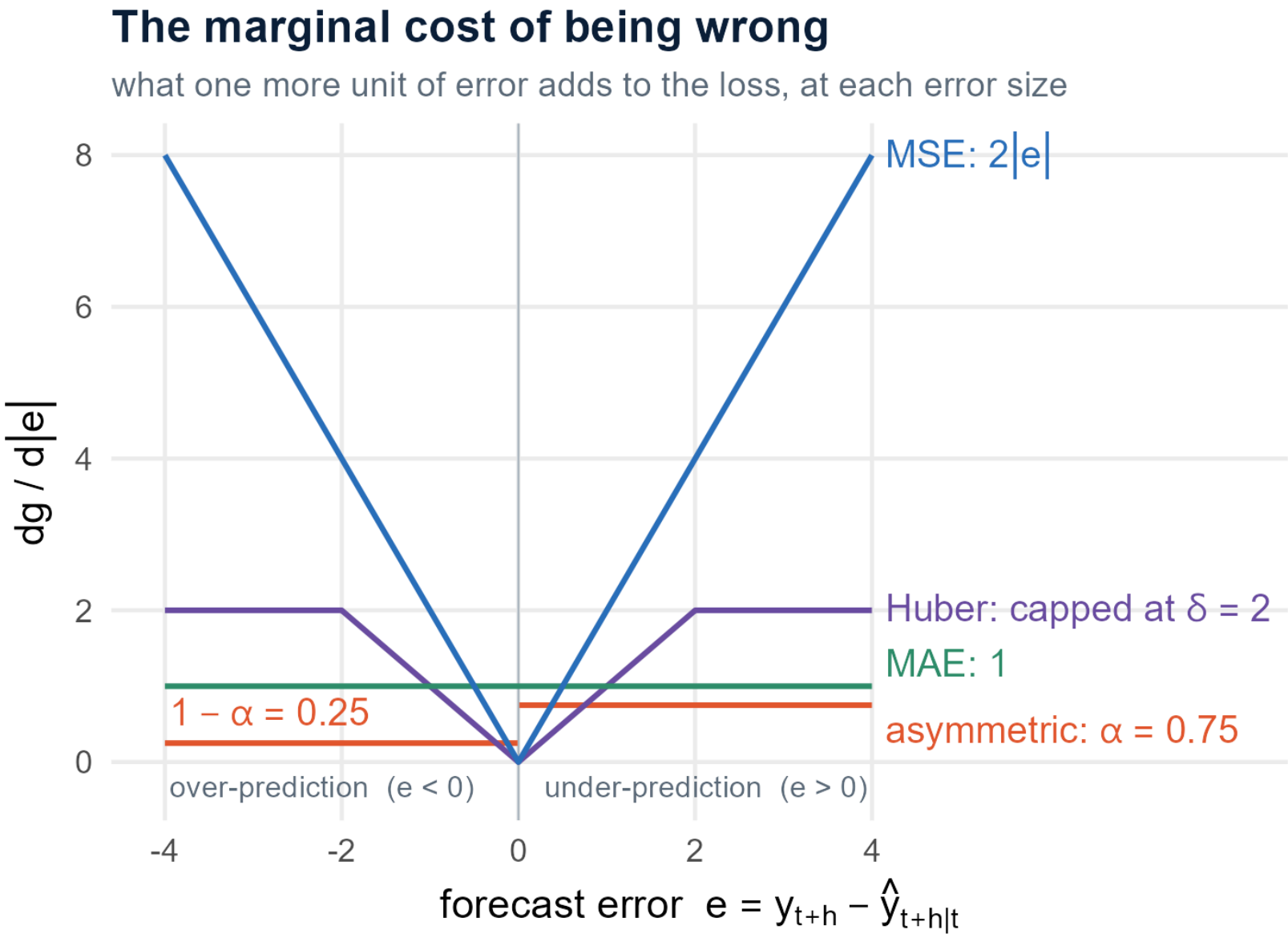
A hospital forecasts daily ER admissions to set its staffing. Too few staff is far more costly than too many. What should the loss look like, and what does the best forecast then represent?

Match the loss to the cost of being wrong

The marginal cost of one more unit of error picks the shape. Ask what each additional unit of miss does to you, then write the loss that charges for it.

Cost of being wrong	Loss	Formula
Explosive per-unit cost each unit of error does more damage than the last	MSE	$\frac{1}{N} \sum_{t=1}^N (y_t - \hat{y}_t)^2$
Constant per-unit cost an error of 10 is exactly twice an error of 5	MAE	$\frac{1}{N} \sum_{t=1}^N y_t - \hat{y}_t $
Quadratic for small errors, linear for large precision in the normal range, robust to outliers	Huber	$g_{\delta}(e) = \begin{cases} \frac{1}{2}e^2 & \text{if } e \leq \delta \\ \delta(e - \frac{1}{2}\delta) & \text{if } e > \delta \end{cases}$
Directional cost one direction costs more than the other	Asymmetric	$\begin{cases} \alpha e & \text{if } e > 0 \text{ (under-prediction)} \\ (1 - \alpha) e & \text{if } e \leq 0 \text{ (over-prediction)} \end{cases}$

N counts test-set observations, not the sample size T . $\delta > 0$ sets where Huber switches from quadratic to linear; $\alpha \in (0, 1)$ is the weight on under-prediction, and $\alpha = 0.5$ is MAE.

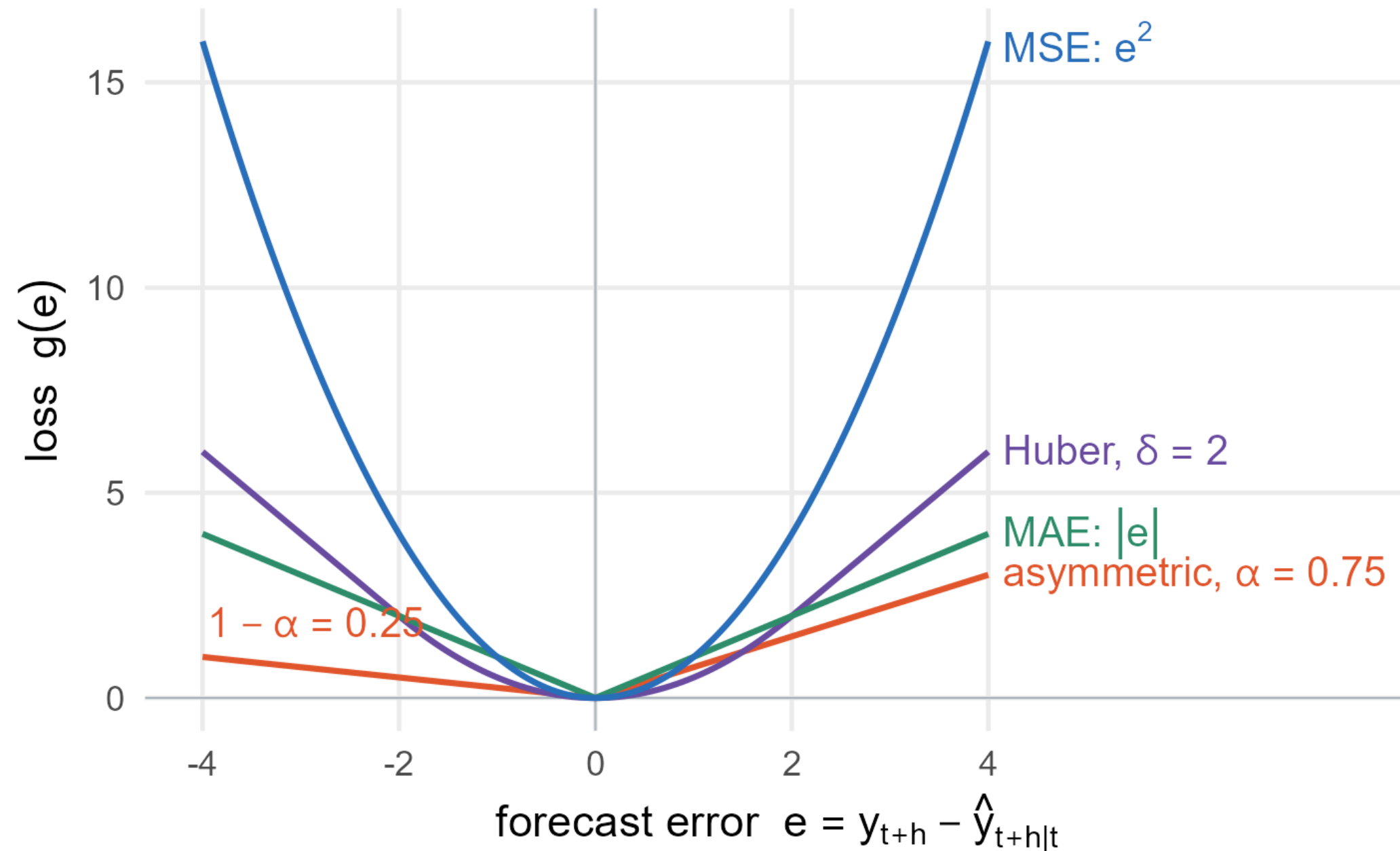


The height of each line is what the next unit of error costs. MSE’s rises without bound; MAE’s is flat; Huber’s rises, then stops at $\delta = 2$; the asymmetric loss charges $\alpha = 0.75$ per unit of under-prediction and 0.25 per unit over.

The shapes

Four loss functions, one axis

$g(e)$ for e from -4 to 4; zero at a perfect forecast, larger as the miss grows



The quadratic explodes. MSE is the cost curve where tail events are catastrophic: off by 10 is four times as bad as off by 5, not twice.

The linear grows steadily. MAE: every unit of error costs the same.

Huber caps marginal influence. A quadratic near zero, a straight line beyond $\delta = 2$: large errors are not costless, but no single outlier can dominate the evaluation.

The asymmetric loss charges more on one side. $\alpha = 0.75$ per unit of under-prediction, 0.25 per unit of over-prediction.

MAPE, $\frac{100}{N} \sum_{t=1}^N \left| \frac{y_t - \hat{y}_t}{y_t} \right|$, is scale-free but undefined at $y_t = 0$ and asymmetric by construction. Rarely the right choice for a macro series.

The loss picks the target

Under MSE, the best point forecast is the	conditional mean
Under MAE	conditional median
Under asymmetric linear loss with weight α	the α -quantile

The Texas operator at $\alpha = 0.9$ forecasts the 90th percentile of demand, not the mean. Symmetric shocks hide the difference; skew them and the loss decides:

```

1 set.seed(2020); eps <- rexp(600) - 1 # skewed right
2 y <- numeric(600)
3 for (t in 2:600) y[t] <- 0.6 * y[t - 1] + eps[t]
4 y_train <- window(ts(y), end = 350)
5 fit <- Arima(y_train, order = c(1, 0, 0))
6 full <- Arima(ts(y), model = fit) # no refit
7 fc_mean <- window(fitted(full), start = 351)
8 fc_median <- fc_mean + median(residuals(fit)) # -0.39

```

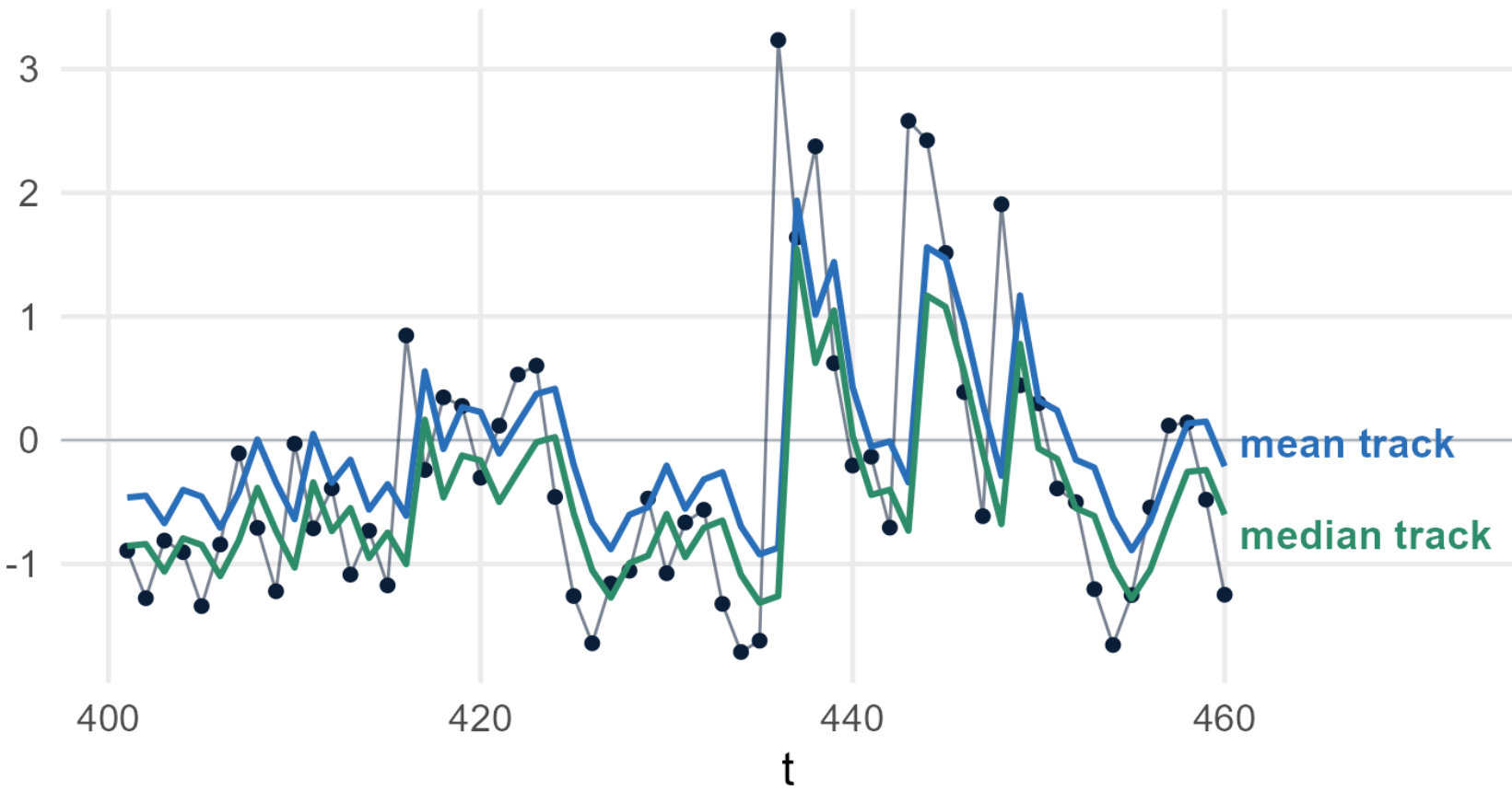
Which forecast wins?

	mean forecast	median forecast
MSE	1.063	1.207
MAE	0.753	0.727

Each wins under the loss it was built for. Not a paradox: the loss doing its job.

Two one-step forecast tracks against the actuals

test set, $t = 401$ to 460 ; dots: the actuals; median track = mean track $- 0.39$



Most shocks are small and negative, a few are large and positive. The mean track leans toward the expensive jumps; the median track sits where most observations land, 0.39 below it.

“A forecast has two parts” called the conditional mean the best guess *under squared-error loss*. This is why the qualifier was there.

Write your own; compare within

The standard losses are starting points, not laws. A loss function is any function of the actual and the forecast that is zero when the forecast is perfect and grows as the forecast gets worse. Beyond that, the shape is yours to choose.

A call centre: under-staffing by one costs \$500 in dropped calls and escalates, over-staffing by one costs \$100 in idle wages. Write the cost, not a textbook metric:

```
1 call_center_loss <- function(actual, forecast) {
2   e <- actual - forecast
3   cost <- ifelse(e > 0,
4                 500 * e + 50 * e^2,      # under-staffed
5                 100 * abs(e))           # over-staffed
6   return(mean(cost))
7 }
```

Score the competing models with `call_center_loss(actuals, fc_A$mean)` and `call_center_loss(actuals, fc_B$mean)`: the winner minimises your actual cost.

Within, never between. Two models are compared under one loss, on one series, over the same N test-set observations. “A wins on MSE, B wins on MAE” is two questions with two answers, and the rule for the next slide.

Next: the GDP forecasts you just watched, scored under each loss. Module 7 asks whether the differences are significant.

The same forecasts, scored under each loss

criterion

AIC

AICc

BIC

HQIC

loss

MSE

MAE

Huber

asymmetric

Reset

scope

all 20 quarters

2008 Q3 to 2009 Q2

δ = 1.00

α = 0.5

AICc picks ARIMA(2,1,0) with drift · same model as AIC and HQIC; BIC picks ARIMA(1,1,0) with drift · its row is marked

MSE = mean(e^2) · the RMSE beneath each cell is its square root, in percent per quarter

e = realised – forecast growth, percent per quarter · all 20 test quarters, 2007 Q1 to 2011 Q4

model	h = 1	h = 2
ARIMA(2,1,0) with drift AIC · AICc · HQIC	0.589 RMSE 0.767	0.704 RMSE 0.839
ARIMA(1,1,0) with drift BIC	0.627 RMSE 0.792	0.804 RMSE 0.897
random walk with drift (benchmark) same at both horizons	0.869 RMSE 0.932	0.869 RMSE 0.932

Under MSE, ARIMA(2,1,0) with drift wins at h = 1 and at h = 2.

inside the band, of 20 (h = 1 · h = 2)	ARIMA(2,1,0) 95%: 18 · 18 80%: 15 · 15	ARIMA(1,1,0) 95%: 17 · 18 80%: 14 · 15	RW with drift 95%: 16 · 17 80%: 13 · 12
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Compare within a loss, never between: an MSE and an MAE are in different units. The loss is chosen before the forecasts, from the cost of being wrong (under-prediction dearer? α above ½), and the forecasts are ranked under it. Growth errors in percent per quarter; parameters frozen at 2006 Q4; 2026-09-30 vintage.